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Q1.(Data Loading & Initial Analysis), Load the Ford Car Dataset using pandas,

Display the first 10 rows and last 5 rows. Check the shape (number of rows and columns) and data types of all columns. Write observations about the dataset structure in comments.

```
import pandas as pd

df = pd.read_csv("ford.csv")

print("First 10 Rows:")

print(df.head(10))

print("\nLast 5 Rows:")

print(df.tail(5))

print("\nShape of Dataset:")

print(df.shape)

print("\nData Types of Columns:")
```

Q2.(Missing & Duplicate Values), Perform a detailed check on the dataset:

Find the total number of missing values in each column. . Identify and remove duplicate rows if any. • Write a summary of your findings and how you handled them in comments

```
import pandas as pd

df = pd.read_csv("ford.csv")

print("Missing Values in Each Column:")

print(df.isnull().sum())

print("\nTotal Missing Values in Dataset:")

print(df.isnull().sum().sum())
```

```
duplicate_count = df.duplicated().sum()
print("\nNumber of Duplicate Rows:")
print(duplicate_count)
df = df.drop_duplicates()
print("\nShape of Dataset After Removing Duplicates:")
print(df.shape)
print("\nDuplicate Rows Remaining:")
print(df.duplicated().sum())
```

Q3.(Statistical Summary)

```
import pandas as pd
df = pd.read_csv("ford.csv")
print("Statistical Summary of Numeric Columns:")
print(df.describe())
print("\nPrice Statistics:")
print("Minimum Price :", df['price'].min())
print("Maximum Price :", df['price'].max())
print("Mean Price    :", df['price'].mean())
print("Median Price  :", df['price'].median())
print("\nMileage Statistics:")
print("Minimum Mileage :", df['mileage'].min())
print("Maximum Mileage :", df['mileage'].max())
print("Mean Mileage    :", df['mileage'].mean())
```

```
print("Median Mileage  :", df['mileage'].median())
print("\nYear Statistics:")
print("Minimum Year :", df['year'].min())
print("Maximum Year :", df['year'].max())
print("Mean Year    :", df['year'].mean())
print("Median Year  :", df['year'].median())
```

Q4.(Histogram of Numeric Features)

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("ford.csv")
df[['price', 'mileage', 'year', 'engineSize', 'mpg']].hist(
    figsize=(15,10),
    bins=20,
    edgecolor='black'
)
plt.tight_layout()
plt.show()
```

Q5.(Count Plots of Categorical Features)

```
plt.figure(figsize=(12,6))
```

```
top_models = df['model'].value_counts().head(10).index
sns.countplot(
    data=df,
    y='model',
    order=top_models
)
plt.title('Top 10 Ford Models')
plt.xlabel('Count')
plt.ylabel('Model')
plt.show()
```

Q6.(Correlation Heatmap)

```
import matplotlib.pyplot as plt
import seaborn as sns
numeric_data = df.select_dtypes(include=['int64', 'float64'])
corr_matrix = numeric_data.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)
```

```
plt.title('Correlation Heatmap of Numeric Features')
plt.show()
price_corr = corr_matrix['price'].sort_values(ascending=False)
print("Correlation with Price:")
print(price_corr)
```

Q7.(Feature Identification)

```
X = df.drop('price', axis=1)
y = df['price']
print("Independent Features:")
print(X.columns.tolist())
print("\nDependent Feature:")
print(y.name)
```

Q8.(Encoding Categorical Variables)

```
categorical_cols = df.select_dtypes(include=['object']).columns
print("Categorical Columns:")
print(categorical_cols)
print("Before Encoding:\n")
print("Model Column:")
print(df['model'].head())
```

```
print("\nTransmission Column:")
print(df['transmission'].head())
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
print("\nDataset Shape Before Encoding:", df.shape)
print("Dataset Shape After Encoding:", df_encoded.shape)
print("\nAfter Encoding:\n")
print("Encoded Model Columns:")
print([col for col in df_encoded.columns if col.startswith('model_')][:5])
print("\nEncoded Transmission Columns:")
print([col for col in df_encoded.columns if col.startswith('transmission_')])
```

Q9.(Feature Scaling)

```
from sklearn.preprocessing import StandardScaler
X = df_encoded.drop('price', axis=1)
y = df_encoded['price']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)
print("First 5 Rows of Scaled Data:")
print(X_scaled_df.head())
```

Q10.(Mini Project - Complete Preprocessing Pipeline) Create a complete preprocessing pipeline for the Ford Car Dataset:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import StandardScaler

df = pd.read_csv("ford.csv")

print("First 5 Rows:")

print(df.head())

print("\nDataset Info:")

print(df.info())

print("\nMissing Values:")

print(df.isnull().sum())

df.dropna(inplace=True)

print("\nDuplicate Rows:", df.duplicated().sum())

df.drop_duplicates(inplace=True)

print("\nDataset Shape After Cleaning:", df.shape)

numeric_cols = ['price', 'mileage', 'year', 'engineSize', 'mpg']

for col in numeric_cols:

    plt.figure(figsize=(6,4))

    plt.hist(df[col], bins=20, edgecolor='black')

    plt.title(f'Histogram of {col}')

    plt.xlabel(col)

    plt.ylabel('Frequency')

    plt.show()
```

```
categorical_cols = ['fuelType', 'transmission', 'model']
for col in categorical_cols:
    plt.figure(figsize=(10,5))
    sns.countplot(x=df[col])
    plt.title(f'Count Plot of {col}')
    plt.xticks(rotation=45)
    plt.show()
plt.figure(figsize=(10,8))
corr_matrix = df.select_dtypes(include=['int64', 'float64']).corr()
sns.heatmap
```